Lesson 1.2

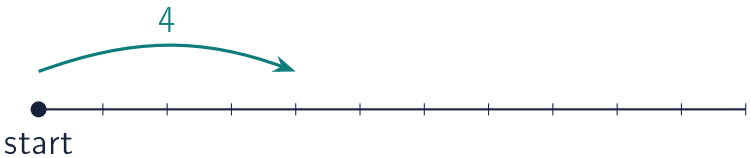
Segment Addition Postulate

Part + part = whole

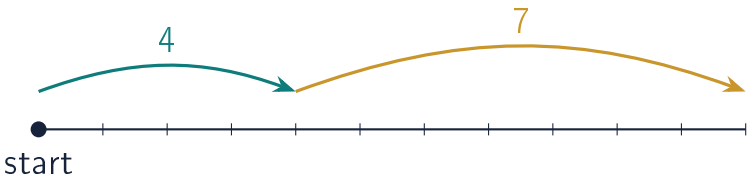
Walk 4 steps, then 7 more



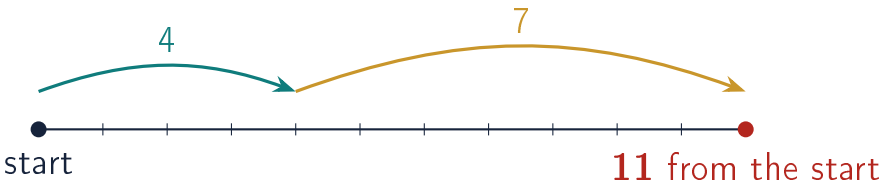
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$4 + 7 = 11$ — you just used today's postulate.

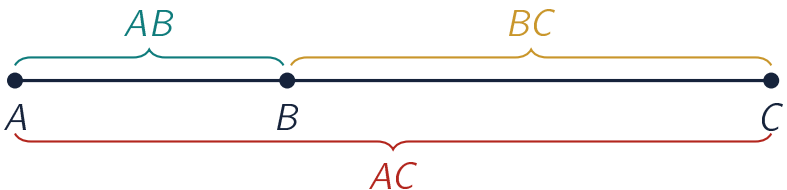
The postulate



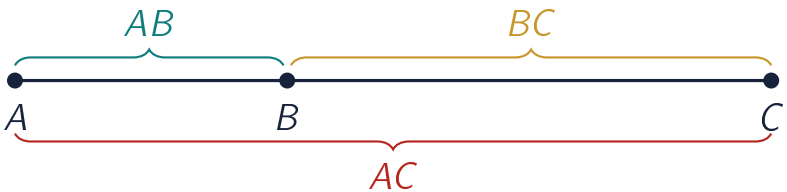
The postulate



The postulate



The postulate



If B is **between** A and C, then $AB + BC = AC$

Simple cases first

Find the whole: add

$AB = 4,\ BC = 7$

Simple cases first

Find the whole: add

$AB = 4,\ BC = 7$

$AC = AB + BC$

Simple cases first

Find the whole: add

$AB = 4,\ BC = 7$

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$AC = 4 + 7 = 11$

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Know the whole and one piece: subtract

$AC = 20,\ AB = 8$

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Find the whole: add

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$AC = 4 + 7 = 11$

Know the whole and one piece: subtract

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$BC = AC - AB$

Simple cases first

Find the whole: add

$AB = 4,\ BC = 7$

$AC = AB + BC$

$AC = 4 + 7 = 11$

Know the whole and one piece: subtract

$AC = 20,\ AB = 8$

$BC = AC - AB$

$BC = 20 - 8 = 12$

With algebra

$AB = x + 3, \quad BC = 2x, \quad AC = 21.$ Find x and BC.

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$(x + 3) + 2x = 21$ part + part = whole

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$AB = x + 3$, $BC = 2x$, $AC = 21$. Find x and BC.

$(x + 3) + 2x = 21$ part + part = whole

$3x + 3 = 21$ combine like terms

With algebra

$AB = x + 3$, $BC = 2x$, $AC = 21$. Find x and BC.

$(x + 3) + 2x = 21$	part + part = whole
$3x + 3 = 21$	combine like terms
$3x = 18$	subtract 3

With algebra

$AB = x + 3$, $BC = 2x$, $AC = 21$. Find x and BC.

$(x + 3) + 2x = 21$	part + part = whole
$3x + 3 = 21$	combine like terms
$3x = 18$	subtract 3
$x = 6$	**don't stop at x!**

With algebra

$AB = x + 3$, $BC = 2x$, $AC = 21$. Find x and BC.

$(x + 3) + 2x = 21$	part + part = whole
$3x + 3 = 21$	combine like terms
$3x = 18$	subtract 3
$x = 6$	**don't stop at x!**
$BC = 2(6) = 12$	the question asked for BC

With algebra

$AB = x + 3$, $BC = 2x$, $AC = 21$. Find x and BC.

$(x + 3) + 2x = 21$	part + part = whole
$3x + 3 = 21$	combine like terms
$3x = 18$	subtract 3
$x = 6$	**don't stop at x!**
$BC = 2(6) = 12$	the question asked for BC

Check: $AB = 9$, $9 + 12 = 21$ ✓

Only when B is between A and C

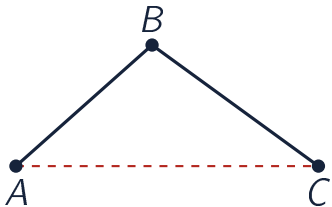


✓ $AB + BC = AC$

Only when B is between A and C



✓ $AB + BC = AC$

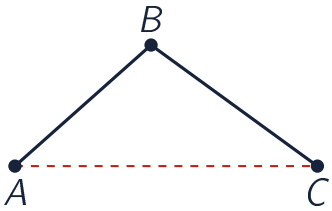


× $AB + BC > AC$

Only when B is between A and C



✓ $AB + BC = AC$



× $AB + BC > AC$

Watch out! If A, B, C make a triangle, the two sides add up to **more** than AC.

Your turn! Pause the video and try it.

Q is between P and R.
$PQ = 3x - 1, \quad QR = x + 5, \quad PR = 24.$

Find x and PQ.

Your turn! Pause the video and try it.

Q is between P and R.
$PQ = 3x - 1, \quad QR = x + 5, \quad PR = 24.$

Find x and PQ.

$$(3x - 1) + (x + 5) = 24 \;\Rightarrow\; 4x + 4 = 24 \;\Rightarrow\; x = 5$$

Your turn! Pause the video and try it.

Q is between P and R.
$PQ = 3x - 1, \quad QR = x + 5, \quad PR = 24.$

Find x and PQ.

$(3x - 1) + (x + 5) = 24 \Rightarrow 4x + 4 = 24 \Rightarrow x = 5$

$PQ = 3(5) - 1 = 14$ check: $14 + 10 = 24$ ✓

Remember

- **Part + part = whole**: $AB + BC = AC$ when B is between A and C.
- Missing piece? **Subtract** from the whole.
- Solve for x — then answer the question they actually asked.

Now practice at your level:

fangmath.com → Geometry → Chapter 1